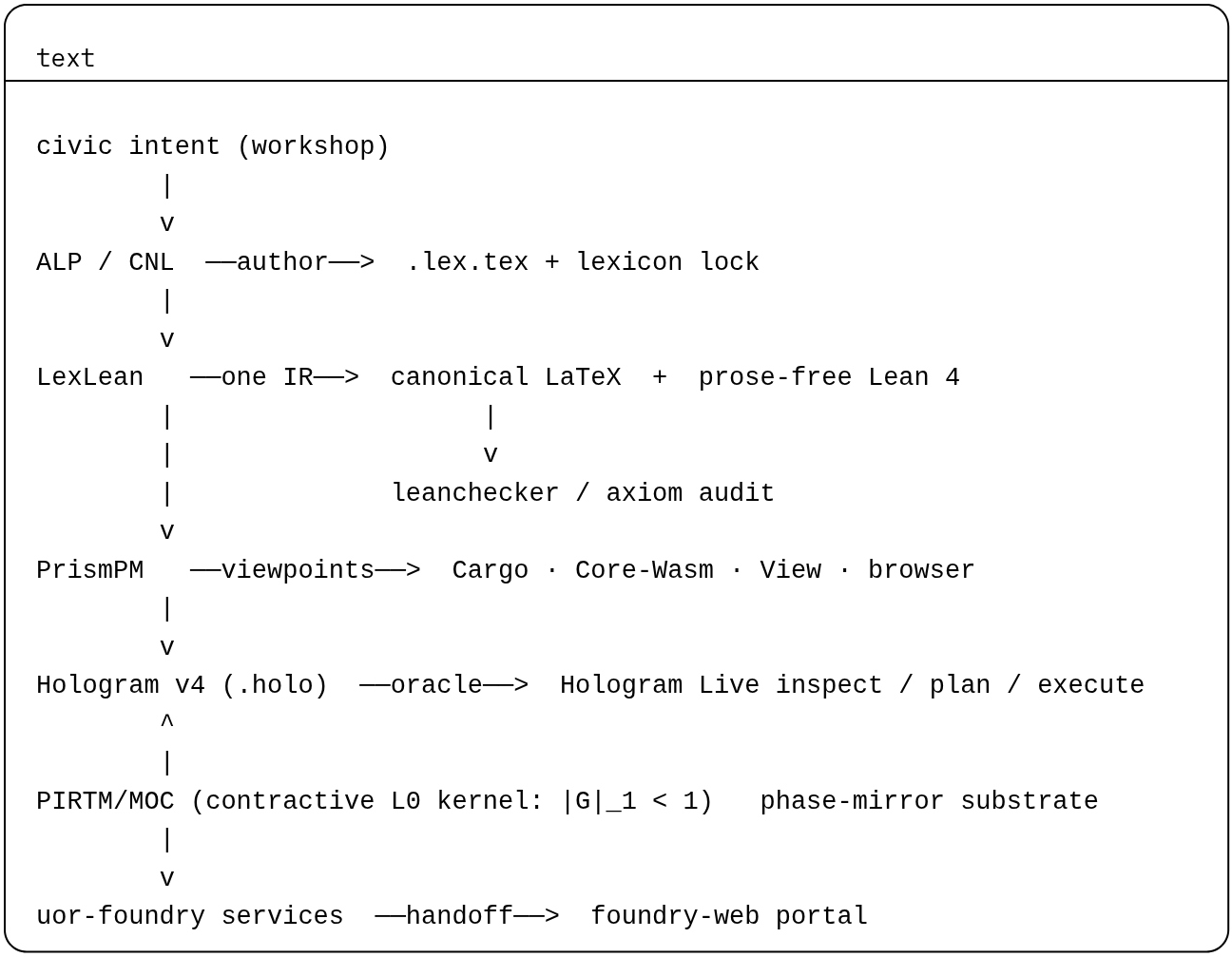
The Foundry is the L0 base-state for uor.foundation: a civic R&D workshop, not a privileged organization identity. Phase-mirror methodology is the rule of the stack. One closed source is reflected into every downstream artifact, and a projection that cannot be replayed from that source is not a Foundry output.

UOR Foundation is one organization on that platform, with the Citizen Gardens operating model and Foundry sites. It is not the platform itself.

The mirror, layer by layer

text



Each arrow is a mirror, not a rewrite. The lower layer may refuse the upper one. It may not invent behavior the upper one did not declare.

ALP/CNL → LexLean

ALP/CNL is the authoring face. CNL here is a closed controlled language: every word, symbol, and control must come from a versioned lexicon package, and every sentence must

parse under a fixed grammar. The public contract does not publish a separate “ALP” repository name. What it does publish is the language LexLean actually compiles.

LexLean is the closed-lexicon LaTeX-to-Lean 4 compiler (`LEXLEAN-SPEC-1`). A `.lex.tex` module is the only source. One typed intermediate representation emits both the canonical LaTeX document and the prose-free Lean program. Verification is pinned (`leanprover/lean4`), replayed with `leanchecker` , and axiom-audited. The working loop is `init → lock → check → build → verify` .

That is the first phase mirror: prose and proof are two faces of one semantic object. A sentence that cannot lower is not yet a Foundry claim.

LexLean → PrismPM

PrismPM is the platform-model compiler. A model is a closed `.lex.tex` source graph, authored in LexLean language 1.1. Prism application behavior is never supplied by handwritten Lean or by target-specific application code.

The v0.3.0 production pipeline is:

1. Author systems, proofs, packaging intent, and typed Views in `.lex.tex` .
2. LexLean emits the semantic snapshot, Lean, LaTeX, maps, and coverage. Observed axioms must match the declared policy.
3. `lean4-prod` exports only named verified roots (LCNF).
4. Generic generators emit the Cargo crate, an import-free `hologram:guest/core-wasm@1` guest, the evaluated View (portable HOLOVIEW plus browser adapter), and the binary archive.
5. An independent Hologram Live oracle validates and executes that archive.

Viewpoints follow ISO/IEC/IEEE 42010:2022 architecture description, with security facets aligned to ISO/IEC 27034. The reference closure is Calculator: `prism-stdlib` , `prism-calculator` , `Calculator.holo` , oracle evidence, and the public `calculator-example` Pages app must all pass the atomic release contract together.

PrismPM is the second mirror: one model, many artifacts, identical identity.

PrismPM → Hologram

Holo/1 is the Prism application profile defined by `prism-stdlib` . It is not a version number. The physical container is a Hologram archive, version 4. Every `.holo` begins with `HOLO\x04\x00` . JSON is never a `.holo` . The canonical model document is the separate

`HOLO\x04\x00`. JSON is never a `.holo`. The canonical model document is the separate `model.prism.json`, a total deterministic projection of the LexLean snapshot, identified by SHA-256.

PrismPM does not take Hologram platform code as a production Cargo dependency. It pins the upstream oracle and generates its own wire codec (`Foundation.Holo.V1.Wire`). The producer extension key is `https://uor.foundation/extension/prismpm-model/v1`. Provenance binds the model blob, LexLean and lean4-prod evidence, generated core/guest/View/browser evidence, dependency commits, and application kappa.

Kappa is the UOR content address: one address per object, so the archive, the guest, and the View can be inspected without trusting the builder. Hologram is the third mirror: the model becomes a portable, resumable object that a foreign oracle can execute.

PIRTM, the contractive kernel

PIRTM/MOC (Phase Mirror / Multiplicity Object Code) is the other L0, on the PhaseMirror side. It is a governed, contractive systems language: Sedona Spine and prime-indexed tensor contractions in the kernel, Lean 4 proofs with no Mathlib, source lowered through MLIR to LLVM. A session runs only after transpile-time receipts and a link-time spectral small-gain check:

$$|G|_1 = \max_j \sum_i |A_{ij}| \cdot \lambda_j < 1 \quad \text{in } \mathbb{Q}$$

held over exact reduced rationals (`PosRat`). Legal entity is Citizen Gardens UNA, d/b/a The Prime Materia Commons.

PIRTM is not a stage inside the PrismPM compiler. It is the stability law the workshop assumes. Phase mirror is not “generate more views.” It is “every reflection must be contractive.” Drift that fails the small-gain gate, or the Foundry attestation boundary (`SIG_GOV_KILL ⇒ UNATTESTED` in the OSCAL/Scopist model, ADR-0110), does not become infrastructure.

Where the Foundry sits

Two public bodies carry the name. They are not the same object.

Body	Role

PhaseMirror/Foundry	Substrate monorepo: Rust engine, Lean ADR scaffold, UCC. Models the Foundation with PrismPM. ADR-0110 binds OSCAL packaging to PrismPM.
UOR-Foundation/uor-foundry	Generic organization platform: org lifecycle, workflows, scoped authority, portal Views. Consumes PrismPM as an SDK, on the Calculator pattern.
foundry-web	Publishes the exact verified portal release and checks the live deployment. It does not keep a second implementation.

No account, organization, or administrator mailbox is pre-seeded. Sites and organizations are created by ordinary workflow. That is the civic claim: the workshop can host the Foundation without the Foundation being hardcoded as root.

Status, stated as the repos state it: PrismPM v0.3.0 is the accepted production-system release. `uor-foundry` reports the producer model and first functional core accepted across the fifteen conformance boundaries (`SB-01` through `IC-01`), with handoff to `foundry-web` digest-bound. `src/Foundry.lex.tex` is still an unaccepted local draft-preview. Its native, Holo, Core-Wasm, browser, and reproducibility gates have to pass before that increment is a deployable claim.

How a workshop move actually travels

1. A civic structure (role, site, control, workflow) is written in the CNL, inside a locked lexicon.
2. LexLean mirrors it into a document humans can read and a Lean term nothing can silently extend.
3. PrismPM mirrors that term into architecture viewpoints and a portable application. No handwritten target code.
4. Hologram seals the result as a v4 archive. Kappa binds it. The Live oracle either executes that archive or rejects it.
5. PIRTM's contractivity law is the admission test underneath: if the reflection is expansive, it does not enter the base-state.
6. `uor-foundry` runs the accepted organization lifecycle. `foundry-web` only publishes what that handoff already authorized.

Dynamic social infrastructure, in this method, is not a diagram of services. It is a closed source that can be re-mirrored into proof, package, guest, view, and archive without a leftover handwritten path.